Sukhwinder Singh

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Summary

Computer Science undergraduate (8.41 CGPA) with internship experience building autonomous AI workflows and deep learning pipelines. Proficient in Java, Spring Boot, microservices architectures, and PyTorch. Solved 400+ algorithmic problems across LeetCode and GeeksforGeeks; seeking Software Development Engineer (SDE) internship roles.

Education

Thapar Institute of Engineering & Technology — Patiala, Punjab
Bachelor of Engineering in Computer Science & Engineering (CGPA: 8.41) — *2023 – 2027*

Government Model Senior Secondary School (GMSSS), Sector 16 — Chandigarh, India
CBSE Class XII – Non-Medical (Percentage: 89.6%) — *2021 – 2023*

Technical Skills

Languages: Java, C++, Python, JavaScript (ES6+), SQL
Backend Frameworks: Spring Boot, Spring Cloud Gateway, Spring Data JPA, Hibernate, Node.js, Express.js
Frontend Development: React.js, Next.js, HTML5, CSS3, Tailwind CSS
Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis
Deep Learning & Vision: PyTorch, Torchvision, CNNs, Vision Transformers (Swin), OpenCV
Tools & Cloud: Docker, Git, GitHub, Postman, AWS (S3, EC2), Maven, Linux/Bash

Experience

ShipMyBuild — May 2026 – July 2026
Software Engineer Intern — *Remote*

- Engineered a 3-state human-in-the-loop workflow, atomically pausing and resuming AI agents for high-impact actions.
- Built an inbound email filtering pipeline via a 3-condition PostgreSQL gate, preventing automated reply loops.
- Implemented Telegram webhook handlers for 6 interactive callbacks with duplicate-request guards and error isolation.

Samsung PRISM Program — Mar 2025 – Sep 2025
Samsung PRISM Research Intern (Virtual) — *Virtual*

- Built a real-time video face anonymization pipeline replacing identities with photo-realistic synthetic faces.
- Integrated face detection, multi-object tracking, and facial embeddings to ensure temporal consistency and privacy.
- Applied face swapping and blending techniques, optimizing models for near real-time GPU inference in Colab.

Academic Projects

CloudShare | *Spring Boot, Spring Cloud, PostgreSQL, Redis, Docker, React.js* [GitHub] [Live Demo]

- Architected a scalable microservices platform with Spring Cloud Gateway routing and stateless RBAC JWT authentication.
- Slashed metadata query latency by 45% via a Redis cache-aside pattern, significantly reducing primary database load.
- Engineered a resilient file ingestion pipeline using Cloudinary chunked multipart streaming to prevent upload memory overflows.

CampusRide | *Spring Boot, Spring Data JPA, PostgreSQL, Redis, Hibernate, React.js* [GitHub] [Live Demo]

- Built a multi-tenant ride reservation platform, eliminating booking race conditions using PostgreSQL row-level locks.
- Sustained sub-120ms p95 latency under peak loads by eliminating Hibernate N+1 query bottlenecks via DTO projections.
- Designed and documented 30+ RESTful APIs with centralized global exception handlers and OpenAPI/Swagger schemas.

CodeE2LFDepth | *PyTorch, Python, Swin-Transformer, CNN, Sintel/ESIM Datasets* [GitHub]

- Engineered a hybrid CNN and windowed Swin Transformer deep learning architecture for event-camera light-field depth estimation.
- Formulated a compound loss coupling spatial gradient penalties with L1 edge losses, yielding +1.8 dB PSNR and +0.06 SSIM gains.
- Accelerated training throughput and cut GPU memory footprint by profiling tensor bottlenecks and vectorizing PyTorch loaders.

Certificates and Leadership

- **Software Engineer Intern Certificate:** ShipMyBuild
- **Samsung PRISM Research Internship Certificate:** Samsung R&D Institute India
- **Java Spring Framework, Spring Boot, Spring AI:** Gen AI Developer Certificate
- **Group Representative (GR):** Batch Representative for Group 2P11 at TIET, Patiala